HOSPITAL READMISSION ANALYSIS

End-to-End Data Analytics Project

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1. Introduction

The objective of this project was to analyze hospital readmission patterns and identify key factors influencing readmission risk. The analysis was conducted in three stages: data exploration and preprocessing in Python, structured aggregation and metric computation in SQL, and interactive visualization in Power BI. This layered approach ensures data accuracy, analytical depth, and clear communication of insights.

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2. Python Analysis (Exploratory Data Analysis & Preparation)

The analysis began in Python to understand the dataset structure and prepare it for structured querying.

Data Cleaning and Inspection

The dataset was loaded and inspected for:

- Missing values
- Incorrect data types
- Duplicate records
- Distribution of categorical variables

Data types were validated to ensure numerical fields such as age, number of inpatient visits, emergency visits, and medication count were correctly formatted.

Categorical fields such as readmitted, diabetes medication, A1C result, and glucose result were standardized to ensure consistent values.

Descriptive Statistics

Basic descriptive statistics were computed to understand:

- Age distribution
- Range of inpatient visits (0–15)
- Range of emergency visits
- Distribution of medication counts

This revealed that most patients had low prior inpatient visits, while very high visit counts were rare, suggesting the need for grouping later in analysis.

Preliminary Grouping & Trend Exploration

Using Python groupby functions, readmission rates were explored across:

- Age groups
- Inpatient visits
- Emergency visits
- Medication counts

Initial trends indicated:

- Higher readmission rates among elderly patients
- Increasing readmission rates with more prior inpatient visits
- Possible association between medication complexity and readmission

Python analysis helped confirm which variables were strong candidates for deeper structured analysis in SQL.

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3. SQL Analysis (Structured Aggregation & Metric Computation)

After validating and understanding the dataset in Python, SQL was used to compute precise aggregated metrics and prepare structured outputs for visualization.

Readmission Rate Calculation

The primary metric calculated in SQL was:

Readmission Rate (%) =
(Number of patients readmitted / Total patients in group) × 100

This was computed across multiple dimensions.

	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	25000	11754	47.02

Age vs Readmission

Patient age was analysed to examine its relationship with hospital readmission rates. The results indicate that readmission rates tend to increase with older age groups, suggesting that elderly patients may face a higher risk of returning to the hospital. This trend may be due to the greater prevalence of chronic conditions and increased healthcare needs among older patients. This confirmed age as a significant predictor.

	age text	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	[80-90)	4516	2239	49.58
2	[70-80)	6837	3336	48.79
3	[60-70)	5913	2770	46.85
4	[40-50)	2532	1127	44.51
5	[50-60)	4452	1966	44.16
6	[90-10...	750	316	42.13

Inpatient Visits vs Readmission

Raw inpatient visits ranged from 0 to 15. SQL grouping revealed:

- Large patient count in 0 visit group
- Very small sample size in extreme values (14–15 visits)
- 100% readmission in some extreme groups due to very small denominators

To prevent misleading interpretation, inpatient visits were grouped into:

0
1–2
3–5
6+

This improved statistical stability and showed a clear upward trend in readmission rate with increasing prior hospitalizations.

	n_inpatient bigint	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
3	2	1909	1234	64.64
4	3	833	586	70.35
5	4	358	273	76.26
6	5	211	171	81.04
7	6	104	94	90.38
8	7	47	40	85.11
9	8	26	24	92.31
10	9	20	17	85.00
11	10	12	11	91.67
12	11	8	8	100.00
13	12	3	3	100.00
14	13	2	1	50.00
15	14	2	2	100.00
16	15	2	2	100.00

Emergency Visits vs Readmission

Patients with a higher number of prior emergency visits tend to show higher readmission rates compared to those with few or no emergency visits. However, categories with very high emergency counts often contain small sample sizes, which can produce unstable percentages and should be interpreted carefully. Overall, the analysis suggests that frequent emergency department usage may be associated with a greater likelihood of

hospital readmission.

	n_emergency bigint	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	0	22272	10037	45.07
2	1	1842	1087	59.01
3	2	525	343	65.33
4	3	167	123	73.65
5	4	83	69	83.13
6	5	40	32	80.00
7	6	18	16	88.89
8	7	18	16	88.89
9	8	6	5	83.33
10	9	6	5	83.33
11	10	6	6	100.00
12	11	3	2	66.67
13	12	2	2	100.00
14	13	1	1	100.00
15	16	2	2	100.00
16	18	3	3	100.00
17	19	2	2	100.00
18	21	1	1	100.00
19	28	1	1	100.00
20	37	1	0	0.00
21	64	1	1	100.00

Medication Count

Patients taking a higher number of medications generally show a higher rate of hospital readmission compared to those taking fewer medications. This may indicate that patients with multiple medications often have more complex or severe health conditions that require ongoing treatment. However, categories with very high medication counts may contain fewer patients, so the percentages should be interpreted with caution.

	n_medications bigint	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	1	66	26	39.39
2	2	83	24	28.92
3	3	198	73	36.87
4	4	275	99	36.00
5	5	419	159	37.95
6	6	631	224	35.50
7	7	828	315	38.04
8	8	1052	463	44.01
9	9	1157	487	42.09
10	10	1268	548	43.22
11	11	1396	664	47.56
12	12	1509	689	45.66
13	13	1459	698	47.84
14	14	1396	673	48.21
15	15	1469	726	49.42
16	16	1379	673	48.80
17	17	1271	629	49.49
18	18	1140	588	51.58
19	19	1021	508	49.76
20	20	933	465	49.84
21	21	804	429	53.36
22	22	728	375	51.51
23	23	659	360	54.63
24	24	518	283	54.63
25	25	468	254	54.27
26	26	407	216	53.07
27	27	380	193	50.79

	n_medications bigint	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
28	28	315	156	49.52
29	29	251	117	46.61
30	30	234	113	48.29
31	31	187	91	48.66
32	32	144	60	41.67
33	33	111	43	38.74
34	34	123	59	47.97
35	35	84	38	45.24
36	36	74	29	39.19
37	37	72	23	31.94
38	38	65	43	66.15
39	39	54	24	44.44
40	40	43	13	30.23
41	41	35	13	37.14
42	42	31	11	35.48
43	43	35	13	37.14
44	44	20	5	25.00
45	45	18	5	27.78
46	46	20	7	35.00
47	47	21	6	28.57
48	48	16	5	31.25
49	49	19	9	47.37
50	50	13	3	23.08
51	51	10	0	0.00
52	52	9	1	11.11
53	53	12	4	33.33
54	54	6	2	33.33

55	55	7	1	14.29
56	56	8	1	12.50
57	57	8	4	50.00
58	58	7	1	14.29
59	59	6	0	0.00
60	60	6	2	33.33
61	61	4	2	50.00
62	62	3	2	66.67
63	63	5	0	0.00
64	64	1	1	100.00
65	65	2	1	50.00
66	68	2	2	100.00
67	69	2	2	100.00
68	72	1	1	100.00
69	75	1	0	0.00
70	79	1	0	0.00

Diabetes Medication

Patients were divided into those on diabetes medication and those not. Differences in readmission rates were observed, suggesting that medication use may reflect underlying disease severity.

A1C Test Results

Patients with abnormal or uncontrolled A1C results demonstrated higher readmission rates compared to those with normal results.

This indicates that poor chronic disease management contributes to readmission risk.

	A1Ctest text	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	no	20938	9935	47.45
2	high	2827	1299	45.95
3	normal	1235	520	42.11

Glucose Results

Patients categorized as High glucose showed higher readmission rates compared to Normal and Low groups.

This further supports the importance of metabolic control.

	glucose_test text	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	high	686	357	52.04
2	normal	689	333	48.33
3	no	23625	11064	46.83

Medication Change

A change in medication was analyzed to determine whether it is associated with hospital readmission. The results indicate that patients who experienced a medication change show a slightly different readmission rate compared to those whose medications remained unchanged. This suggests that medication adjustments during treatment may be related to patient condition and could influence the likelihood of readmission.

	change text	total_patients bigint	total_readmitted bigint	readmission_rate_percentage numeric
1	no	13497	6077	45.02
2	yes	11503	5677	49.35

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4. Power BI Dashboard (Visualization & Interpretation)

Power BI was used to transform structured SQL outputs into interactive visualizations.

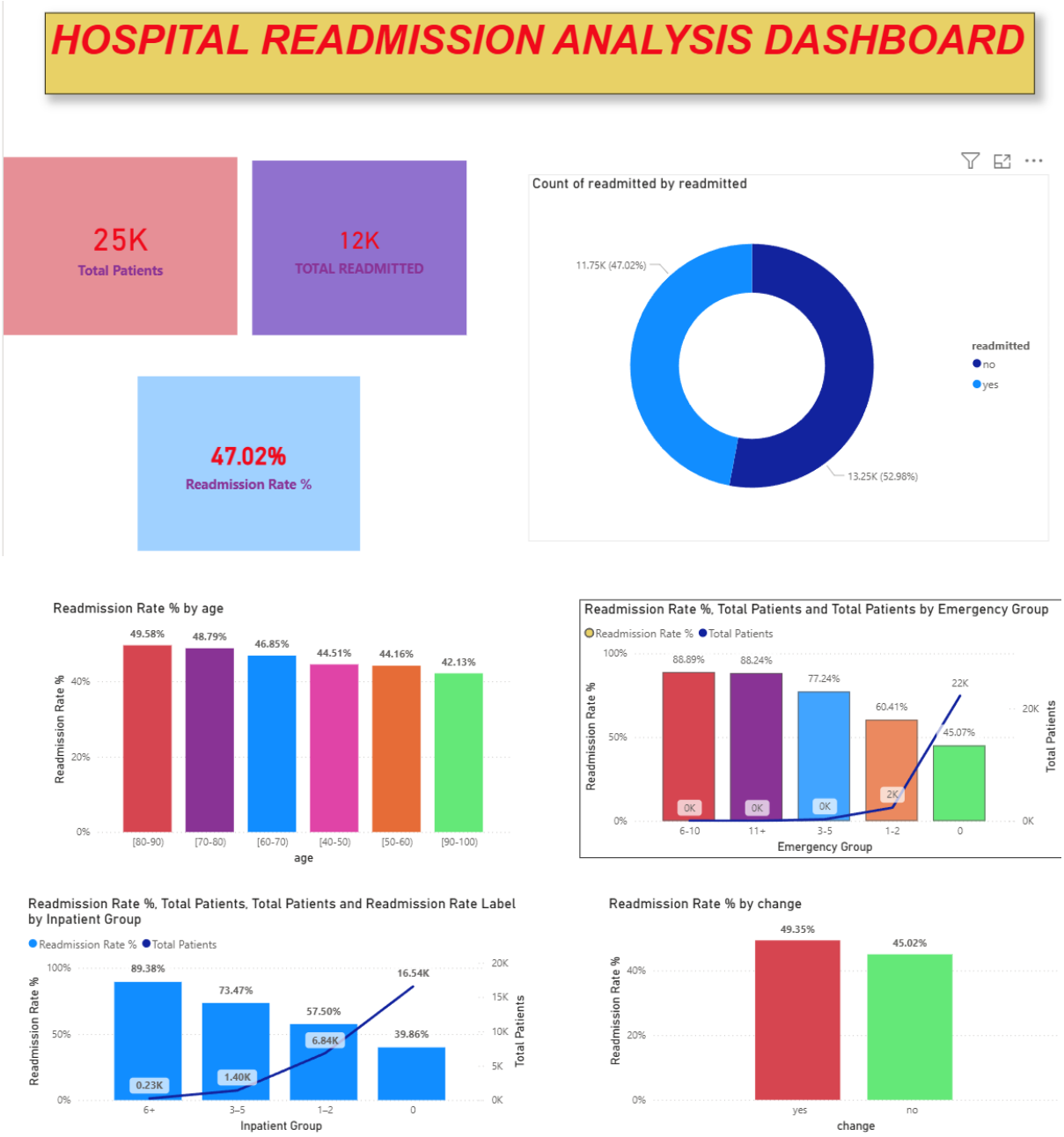
Nine key visuals were developed:

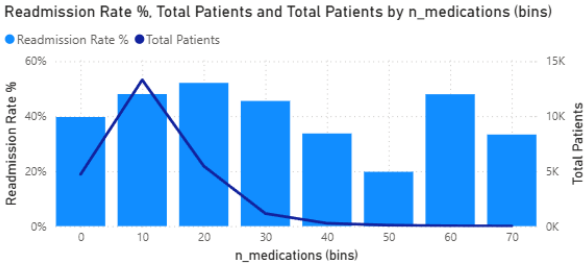
- Age vs Readmission Rate (Line Chart)
- Inpatient Group vs Readmission Rate (Bar Chart)
- Emergency Group vs Readmission Rate
- Medication Count vs Readmission Rate
- Diabetes Medication vs Readmission
- A1C Result vs Readmission
- Glucose Result vs Readmission
- Medication Change vs Readmission
- Overall Readmission Proportion (Donut Chart)

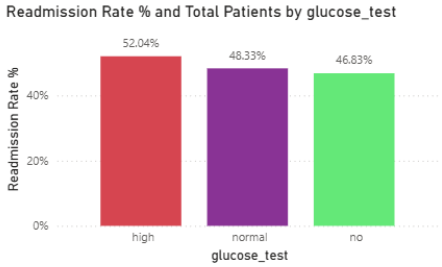
Numeric variables such as inpatient and emergency visits were grouped to avoid instability caused by small sample sizes at extreme values.

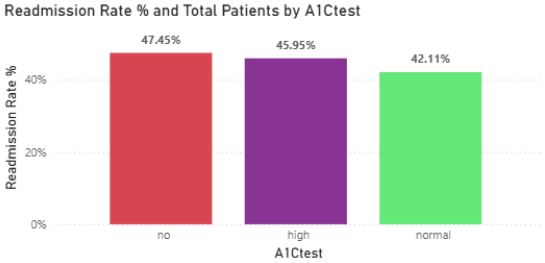
Charts were formatted to display percentage data labels for clarity. In cases where large groups had lower percentages but higher absolute counts (e.g., 0 inpatient visits), interpretation focused on rate-based comparison rather than total counts.

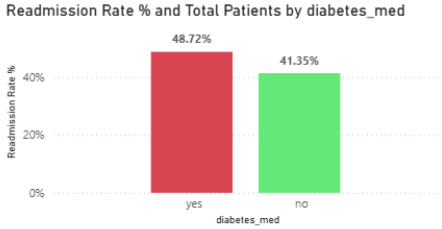
The dashboard allows clear comparison of risk across categories and highlights high-risk patient segments.











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5. Key Insights

The analysis across Python, SQL, and Power BI revealed several important patterns related to hospital readmissions. Patients with a higher number of previous inpatient and emergency visits generally showed higher readmission rates, suggesting that prior hospital utilization is a strong indicator of future readmission risk. Similarly, patients taking a larger number of medications also tended to have higher readmission percentages, which may reflect more complex medical conditions requiring intensive treatment.

Age was another significant factor observed in the analysis, with older patient groups showing higher readmission rates compared to younger groups. Additionally, clinical indicators such as glucose test results and A1C test results showed variation in readmission rates, suggesting that patients with abnormal or unmanaged test results may be at greater risk. Medication changes also showed a slight relationship with readmission, indicating that treatment adjustments during hospitalization may reflect underlying patient health conditions.

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6. Conclusion

Overall, the findings suggest that hospital readmission is influenced by a combination of factors including prior healthcare utilization, patient age, medication complexity, and clinical test outcomes. Identifying these patterns can help healthcare providers better recognize high-risk patients and potentially develop strategies to reduce avoidable readmissions and improve patient care outcomes.